Taiyu Zhang

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SUMMARY

PhD candidate at KU Leuven Computer Science. Hands-on across the language-model lifecycle: from-scratch pretraining (40 GB cleaned Chinese corpus, GPT-2 scale, shipped in the iFlyTek input method), large-model distillation (LingBot video diffusion, 50 min → < 4 min), autoregressive sequence modeling (text-to-motion), and LLM-agent systems (IVA 2024 first author; 1.3k+ GitHub stars).

EXPERIENCE

Part-time Research Engineer — Moxin Technology Mar 2026 – May 2026

- On Huawei Ascend servers, performed self-forcing distillation from a 14B teacher to a 14B student; added camera-trajectory control conditioning on top.
- Applied 4-step few-step distillation to the student on a 1000-frame long-video pretrained model.

NLP Researcher — iFlyTek 2021 – 2022

- Multiple national-level R&D projects; main owner of the multilingual semantic-understanding component of the iFlyTek Digital Human — data curation, model training, and evaluation.
- Deployed in the 2022 Beijing Winter Olympics and other state-level events.
- Long-sentence-completion GPT pretraining for the iFlyTek input method (see Selected Projects).

SELECTED PROJECTS

iFlyTek IME Long-Sentence Completion — *GPT pretraining from scratch* 2021 – 2022

- Built a 400 GB → 40 GB pretraining-data pipeline: deduplication, language and quality filtering, topic balancing, formatting normalization.
- Pretrained a GPT-style language model from scratch on the cleaned 40 GB Chinese corpus.
- Shipped in the iFlyTek input method, powering long-text completion and poetry-creation features in production.

LingBot Distillation — *14B video diffusion → 4-step student on Ascend* 2026.03 – 2026.05

- Reproduced LingBot's distillation on Huawei Ascend (CANN); teacher: camera-trajectory control + 1000-frame long-video coverage. 40–50-step teacher → 4-step student; > 50-min generation → < 4 min.
- Resolved self-forcing memory bottleneck with two-pass forward + activation recompute (gradient-accumulation style); proposed a new ODE-initialization that uses training data to init student directly from teacher, removing explicit ODE-pair generation and stabilising distillation.

Focus Agent — *First author, IVA 2024* 2023 – 2024

- LLM-driven virtual focus group: each participant is an agent with persona, reasoning style, and memory; moderator steers toward an interview goal.
- Published at the 24th ACM International Conference on Intelligent Virtual Agents.

Claw-AI-Lab — *Co-developer* 2025 – Present

- Multi-agent autonomous research platform: one prompt → paper, code, figures, and experiment logs. 1.3k+ GitHub stars (Claw-AI-Lab/Claw-AI-Lab).
- Contributed across the multi-agent scheduler, the local-execution harness, and the real-time monitoring dashboard.

3D Character Animation Generation — *Autoregressive token models + distillation* 2023 – 2025

- Tracked both branches of text-to-motion: discrete-token autoregressive (T2M-GPT, Momask, MotionGPT) and diffusion-based (MDM and successors).
- Swapped VQ-VAE for VQGAN as the motion tokenizer (improved sequence reconstruction); applied few-step distillation to a diffusion motion generator (preserved FID).

EDUCATION

Ph.D., Computer Science — KU Leuven 2022 – Present

Intelligent virtual agents in 3D environments. Advisor: Prof. Adalberto L. Simeone.

M.S., Electronic Engineering — Tsinghua University 2018 – 2021

Advisor: Prof. Xiaohong Ma.

B.S., Electronic Engineering & Information Science — USTC 2014 – 2018

Advisor: Prof. Jian Tang.

SELECTED PUBLICATIONS

Focus Agent: LLM-Powered Virtual Focus Group.
Zhang, T., et al. *Proceedings of the 24th ACM International Conference on Intelligent Virtual Agents (IVA)*, 2024.

SKILLS

Languages	Python · C++ · Swift · TypeScript · C#
ML / Systems	PyTorch · Pretraining · Distillation · Tokenization (VQ-VAE / VQGAN) · Distributed training · Ascend / CANN
Modeling	Transformer / GPT · Diffusion · Autoregressive sequence modeling · Multi-agent orchestration
Spoken	English (fluent) · Chinese (native)